EEE407 Tutorial 6b

1. A horizontal-axis wind turbine with rotor 20 meters in diameter is 30-% efficient in 10 m/s winds at 1 atmosphere of pressure and 15◦C.

   a. How much power would it produce in those winds?

   b. Estimate the air density on a 2500-m mountaintop at 10◦C.

   c. Estimate the power the turbine would produce on that mountain with the same windspeed assuming its efficiency is not affected by air density.

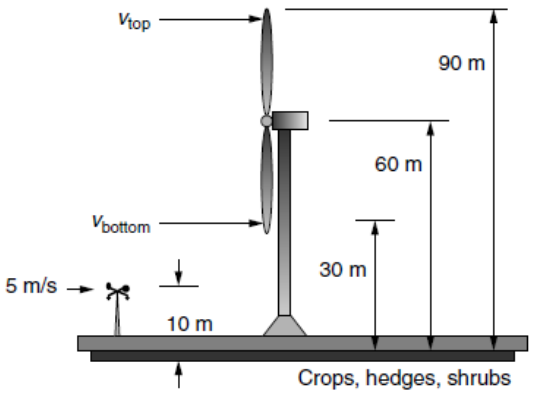
Temperature (°C)	Temperature (°F)	Density (kg/m^3)	Density Ratio (K_T)	Altitude (meters)	Altitude (feet)	Pressure (atm)	Pressure Ratio (K_A)
−15	5.0	1.368	1.12	0	0	1	1
−10	14.0	1.342	1.10	200	656	0.977	0.977
−5	23.0	1.317	1.07	400	1312	0.954	0.954
0	32.0	1.293	1.05	600	1968	0.931	0.931
5	41.0	1.269	1.04	800	2625	0.910	0.910
10	50.0	1.247	1.02	1000	3281	0.888	0.888
15	**59.0**	**1.225**	1.00	1200	3937	0.868	0.868
20	68.0	1.204	0.98	1400	4593	0.847	0.847
25	77.0	1.184	0.97	1600	5249	0.827	0.827
30	86.0	1.165	0.95	1800	5905	0.808	0.808
35	95.0	1.146	0.94	2000	6562	0.789	0.789
40	104.0	1.127	0.92	2200	7218	0.771	0.771

2. An anemometer mounted at a height of 10 m above a surface with crops, hedges and shrubs, shows a windspeed of 5 m/s. Assuming 15◦C and 1 atm pressure, determine the following for a wind turbine with hub height 60 m and rotor diameter of 60 m:

   a. Estimate the windspeed and the specific power in the wind (W/m^2) at the lowest point that a rotor blade reaches.

   b. Repeat (a) at the highest point at which the blade falls.

   c. Compare the ratio of wind power at the two elevations using results of (a) and (b).



Terrain Characteristics	Friction Coefficient α
Smooth hard ground, calm water	0.10
Tall grass on level ground	0.15
High crops, hedges and shrubs	0.20
Wooded countryside, many trees	0.25
Small town with trees and shrubs	0.30
Large city with tall buildings	0.40

3. Estimate the average power in the wind at a height of 50 m when the windspeed at 10 m averages 6 m/s. Assume Rayleigh statistics, a standard friction coefficient $\alpha = 1/7$, and standard air density $\rho = 1.225$ kg/m3.

4. Suppose an anemometer mounted at a height of 10-m on a level field with tall grass shows an average windspeed of 6 m/s.

   a. Assuming Rayleigh statistics and standard conditions (15°C, 1 atm), estimate the average wind power (W/m^2) at a height of 80 m.

   b. Suppose a 1300-kW wind turbine with 60-m rotor diameter is located in those 80 m winds. Estimate the annual energy delivered (kWh/yr) if you assume the turbine has an overall efficiency of 30%.

   c. What would the turbine's capacity factor be?

5. Consider the design of a home-built wind turbine using a 350-W automobile dc generator. The goal is to deliver 70 kWh in a 30-day month.

   a. What capacity factor would be needed for the machine?

   b. If the average wind speed is 5 m/s, and Rayleigh statistics apply, what should the rotor diameter be?

   c. How fast would the wind have to blow to cause the turbine to put out its full 0.35 kW if the machine is 20% efficient at that point?

6. A 750-kW wind turbine with 45-m blade diameter operates in a wind regime that is well characterized by Rayleigh statistics with average windspeed equal to 7 m/s.
   What is the average efficiency of this machine?